

Mobina Pournemat

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📍 Mountain View, California

OVERVIEW

I am a third-year Computer Science Ph.D. student at the University of Maryland, College Park, advised by Prof. Soheil Feizi. My research centers on reasoning in large language models — in particular **efficient inference and reasoning**, **reasoning faithfulness**, and the **evaluation** of what these models can and cannot do reliably. My recent work includes a benchmark for probabilistic reasoning under uncertainty (EMNLP 2026, with Meta AI), methods for lowering the cost of long chain-of-thought reasoning, and a study of LLM answer consistency beyond accuracy. I also have experience in multimodal generative models and computer vision, and prior industry experience in applied machine learning and data science.

EDUCATION

- **University of Maryland, College Park** Maryland, USA
Ph.D. in Computer Science; Advised by Prof. Soheil Feizi; GPA: 4/4 Aug. 2024 - Present
- **Sharif University of Technology** Tehran, Iran
B.Sc. in Computer Engineering Sep. 2018 - Feb. 2023

RESEARCH EXPERIENCE

- **University of Maryland, College Park** Maryland, USA
Research Assistant; Advisor: Prof. Soheil Feizi Aug. 2024 - Present
 - **Efficient Reasoning through Cross-Model Latent Knowledge Transfer (in progress):**
Reducing long chain-of-thought inference cost by transferring reasoning knowledge across models in latent space.
 - **Exploring LLMs' Probabilistic Reasoning Capabilities (EMNLP 2026, with Meta AI):**
Built a comprehensive benchmark for reasoning under uncertainty and uncovered systematic failure modes.
 - **Evaluating LLM Reliability Beyond Accuracy:**
Quantified how consistently LLMs answer semantically identical questions.
 - **Fine-Grained Instruction Control for Image Editing (CVPR 2026):**
Contributed to SliderEdit, a continuous, per-instruction control interface for instruction-based image editing.
- **Institute of Science and Technology Austria (ISTA)** Klosterneuburg, Austria
Machine Learning Intern; Supervisor: Prof. Johann Danzl Nov. 2023 - Jul. 2024
 - **3D Semantic Segmentation, Microscopy Image Analysis:**
Developed an efficient pipeline for automated semantic segmentation of brain microscopy images using *Segment Anything*.
- **Sharif University of Technology** Tehran, Iran
NLP Research Assistant; Advisor: Dr. Ehsaneddin Asgari Jul. 2022 - Oct. 2023
 - **Persian Syntactic and Semantic Error Correction Tool:**
Developed an auto-correction tool for Persian text, combining a BERT-based language model with rule-based modules.
- **Sharif University of Technology** Tehran, Iran
Computer Vision Research Assistant; Advisor: Dr. Mohammad Hossein Rohban Sep. 2021 - Jul. 2022
 - **Anomaly Detection with Autoencoders; Semantic Segmentation:**
Benchmarked reconstruction- and representation-based methods on standard vision datasets.

PUBLICATIONS

- **Reasoning Under Uncertainty: Exploring Probabilistic Reasoning Capabilities of LLMs**
M Pournemat, K Rezaei, G Sriramanan, A Zarei, H Eghbalzadeh, Y Wang, J Fu, F Peng, Z Wang, X Feng, N Noorshams, S Feizi - (EMNLP 2026, Main Conference)
- **SliderEdit: Continuous Image Editing with Fine-Grained Instruction Control**
A Zarei, S Basu, M Pournemat, S Nag, R Rossi, S Feizi - (CVPR 2026, Oral Presentation)
- **Same Question, Different Answers: Evaluating LLM Reliability Beyond Accuracy**
K Faghhi, Y Cheng, S Saha, M Pournemat, A Gerami, S Feizi - (Under Review)

WORK EXPERIENCE

- **Behin Bazaar Marketing Agency** Tehran, Iran
Data Analyst, Business Intelligence *Apr. 2023 - Oct. 2023*
 - **Data Analysis, Database Design, Data Preprocessing, Visualization, Reporting:**
Designed the analytics database and delivered SQL and Power BI dashboards used for campaign decisions.
- **AIMedic** Tehran, Iran
Machine Learning for Healthcare Intern *Jul. 2021 - Sept. 2021*
 - **Data Science, Classification, Semantic Segmentation:**
Built and evaluated classification and segmentation models for medical imaging data.

SELECTED PROJECTS

- **Quantifying Attributional Bias in LLM Toxicity Detection** (Nov. 2025):
Measuring how toxicity scores for identical comments change when attributed to different author identities.
- **Benchmarking Multi-Stage Vulnerability Reasoning in LLMs** (Oct. 2025):
Evaluating the effect of explicit reasoning and post-verification on LLM performance in vulnerability detection.
- **Explainable Detection of Online Sexism** (May 2022):
SemEval-2023, Task 10: hierarchical classification and explanation of sexist content.
- **Recommender System Design** (May 2021):
Content-based, collaborative filtering, and hybrid recommenders with offline evaluation of ranking quality.

TEACHING ASSISTANT EXPERIENCE

- **Introduction to Artificial Intelligence:** Fall 2024
- **Data Structures and Algorithms:** Fall 2022
- **Natural Language Processing:** Spring 2022
- **Computer Simulation:** Fall 2021
- **Probability and Statistics:** Spring 2021

RELEVANT COURSEWORK

- **Natural Language Processing:** 4/4 (A)
- **Long-Context Language Models:** 4/4 (A)
- **Multimodal Foundation Models:** 4/4 (A)
- **Large Language Models Security and Privacy:** 4/4 (A)
- **Advanced Numerical Optimization:** 4/4 (A)
- **Artificial Intelligence:** 20/20
- **Data Structures and Algorithms:** 19.6/20
- **Stanford CS229 - Machine Learning:** online, audited
- **Natural Language Processing Specialization:** Coursera

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, SQL, JavaScript, PHP
- **Machine Learning Libraries:** PyTorch, Hugging Face Transformers, TensorFlow Keras, NumPy, Pandas, scikit-learn
- **Web Development & Database:** HTML/CSS, jQuery, Bootstrap, MySQL, PostgreSQL, SQL Server
- **Miscellaneous:** Microsoft Office, Power BI, Docker, Git, Linux, L^AT_EX

LANGUAGES

Persian: native **English:** proficient **Spanish:** low proficiency **Turkish:** low proficiency